



From lecture session 9: We understood how the molecular structure of the water impacts its thermal properties and scratch the surface of its connection to the Earth's climate and weather patterns.

Lecture Session 10

1. Salinity;
2. Density of seawater (Ocean Layers):
Temperature, Salinity, and Pressure

“Icebreaker”

This video shows an simpler version of the experiment we did in class.

Main take-away: Ocean is stratified (or layered). The density stratification constrain vertical motions. In this case, making the heat-exchange between the water and ice, less efficient.



An aerial photograph of a tropical coastline. A narrow, white sandy beach runs diagonally from the bottom left towards the top right. To the left of the beach is the deep, dark blue ocean. To the right is a shallow lagoon with vibrant turquoise water, revealing the intricate patterns of a coral reef. The overall scene is serene and natural.

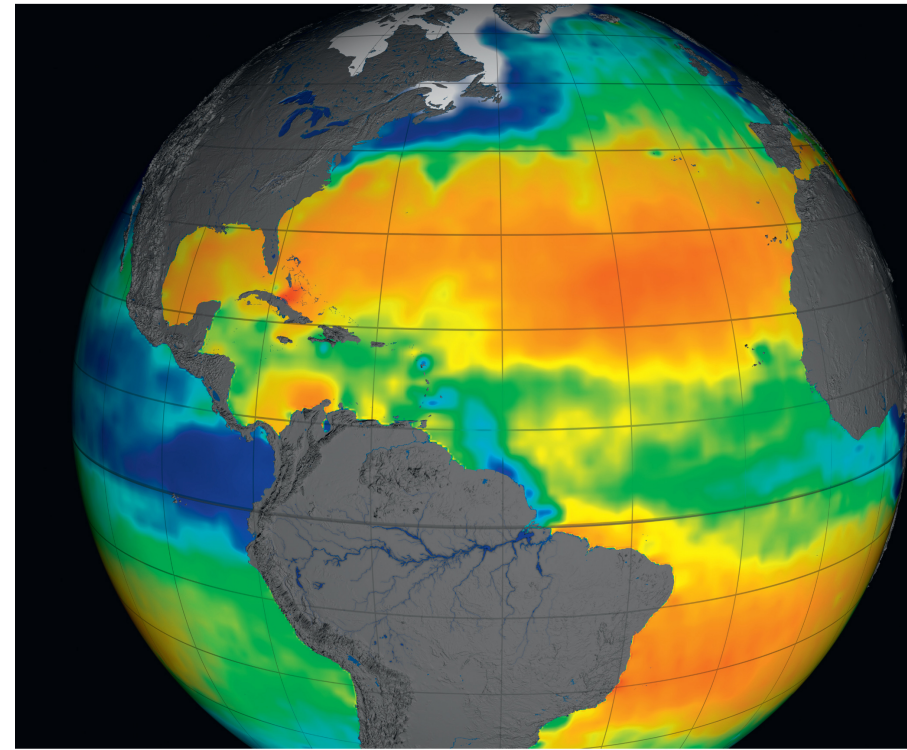
Salinity

As a result of the polar characteristic of the water molecule, the ocean can dissolve a lot of salt. Basically, water breaks the ionic bonds (connection where atoms exchange electrons making salts polar) that hold salt crystals together.

Salts originate in rocks and volcanic activity, accumulating in the ocean (similarly to sediments). However, in contrast to sediments, they are dissolved, and play a more important role in water properties and circulation.

Key Concept: Salinity = grams of salt dissolved in 1 kg of water (aka parts per thousand ‰)

Sea Surface Salinities from satellites measurements.
Red colors are higher salinities.



An aerial photograph of a tropical coastline. The top half of the image shows deep blue ocean water. Below this is a shallow reef flat with vibrant turquoise and green water, showing intricate patterns of coral and sand. A narrow, light-colored sandy beach runs diagonally across the middle. The bottom half of the image shows a vast, flat expanse of green, likely a mangrove forest or a large lagoon. The overall scene is serene and natural.

Density of Seawater

Density = mass/volume

Density mainly depends on Temperature (heat), but also on Salinity, and Pressure

Temperature impact on density: Thermal expansion and contraction of water

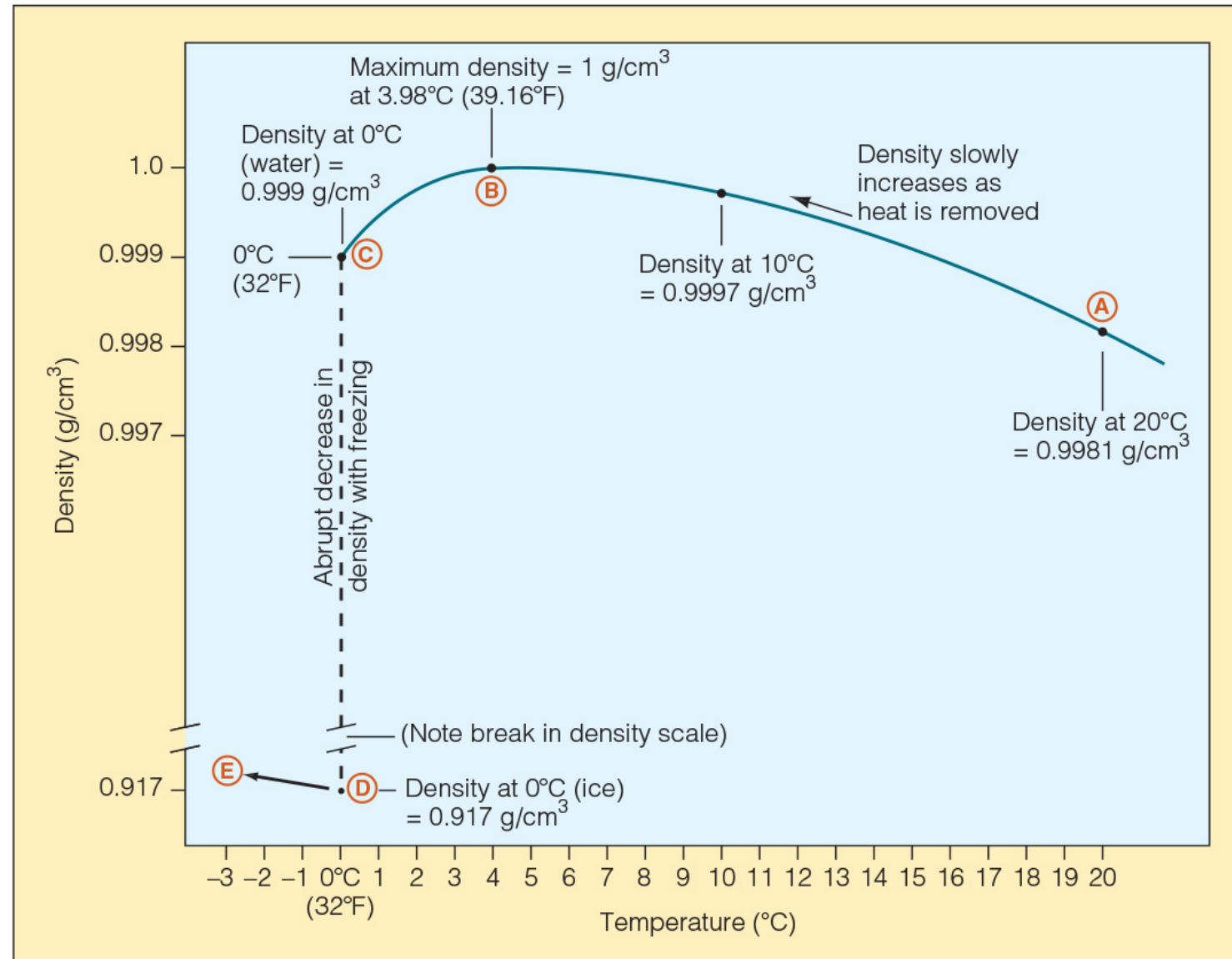


Figure 6.6 The relationship of density and temperature for pure water. Note that points C and D both represent 0°C (32°F) but show different densities, and thus different states of water. Ice floats because the density of ice is lower than the density of liquid water.

Temperature impact on density: Thermal expansion and contraction of water

Usually, molecules tend to occupy more space (volume) if they are vibrating more vigorously.

At temperatures lower than 4°C, the reorganization of the molecules make the water expand.

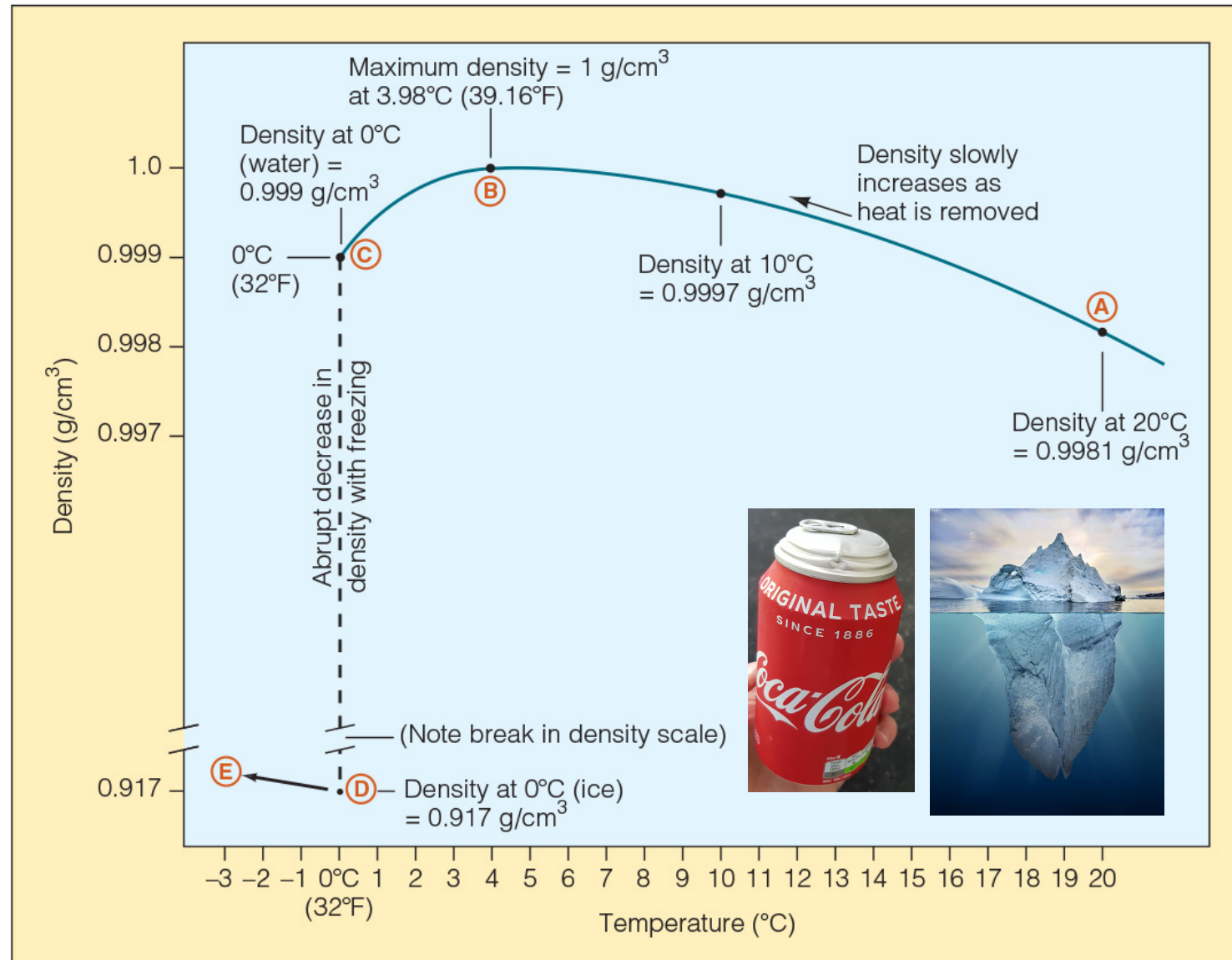


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When we account for salinity: More salt, more mass, and higher density

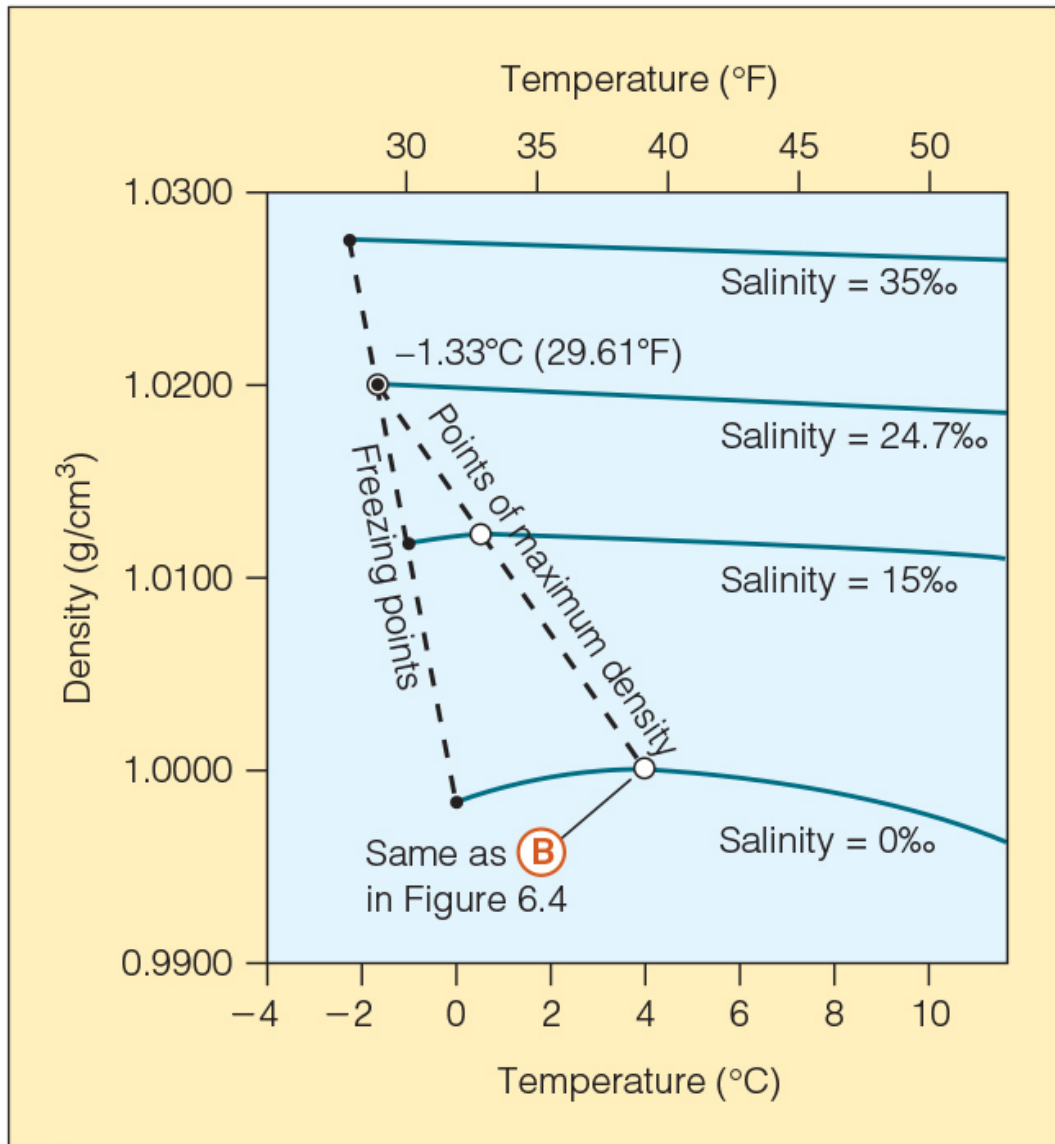
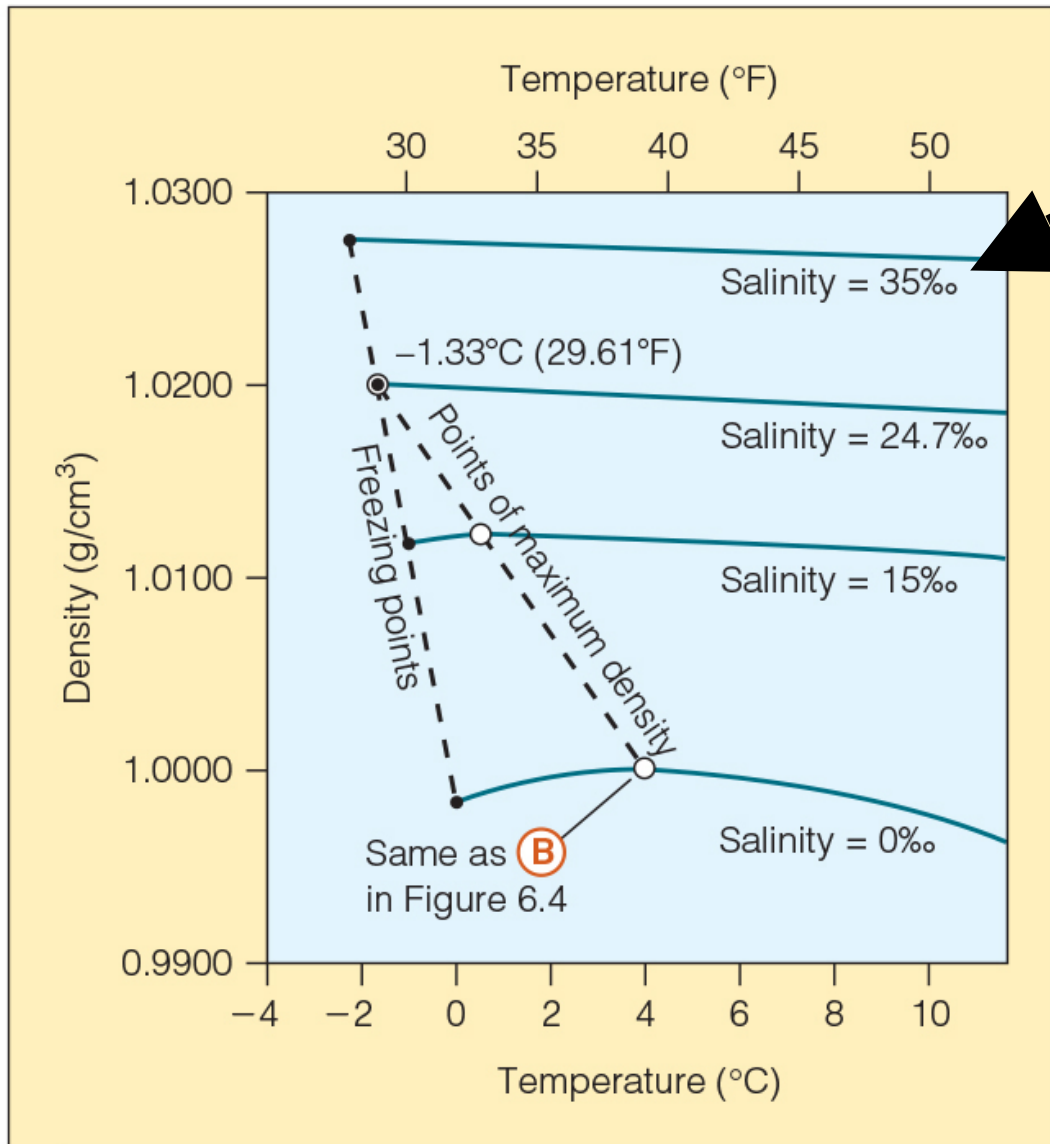


Figure 6.12 The dependence of freezing temperature and temperature of maximum density upon salinity (salt content). As we saw in Figure 6.6, pure water is densest at 3.98°C (39.2°F) (point B), and its freezing point is 0°C (32°F). Seawater with 15‰ salinity is densest at 0.73°C (33.31°F), and its freezing point is -0.80°C (30.56°F). The temperature of maximum density and the freezing point coincide at -1.33°C (29.61°F) in seawater with a salinity of 24.7‰. At salinities greater than 24.7‰ the density of water always decreases as temperature increases. Note that the symbol ‰ represents parts per thousand, so 2.47‰; 24.7‰.

When we account for salinity: More salt, more mass, and higher density



Typical ocean salinity

Salts make the water more “resistant” to freezing and evaporation, and the ions affect the restructuration of the molecules in low temperatures.

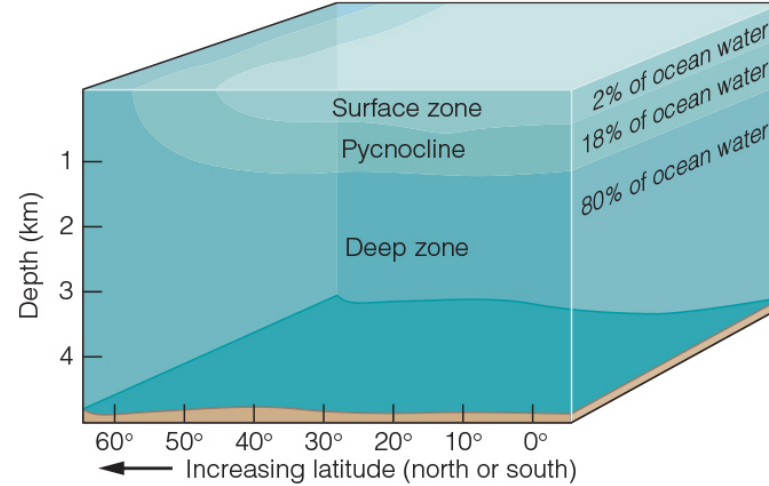
Pressure: It is worth mentioning that pressure pushes water molecules together, making it more compact.

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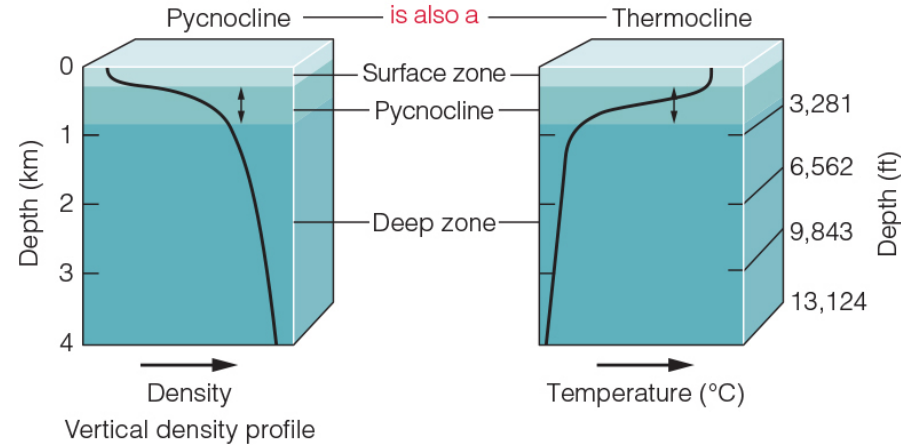
Ocean stratification: Yes layers!

Like the Earth, lighter layers fluctuate over denser layers.

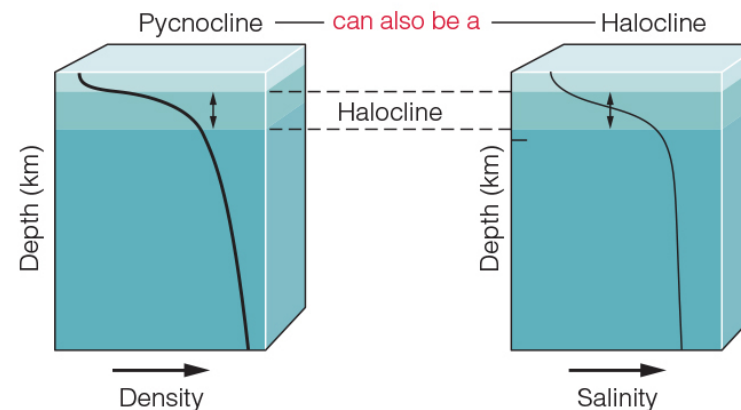
This layered (or stratified) structure constrain vertical motions in the ocean. Ocean currents tend to move more horizontally. It also impacts how light and sound move in the water (refraction).



- a** In most of the ocean, a surface zone (or mixed layer) of relatively warm, low-density water overlies a layer called the *pycnocline*. Density increases rapidly with depth in the pycnocline. Below the pycnocline lies the deep zone of cold, dense water—about 80% of total ocean volume.



- b** The rapid density increase in the pycnocline is mainly due to a decrease in temperature with depth in this area—the *thermocline*.



- c** In some regions, especially in shallow water near rivers, a pycnocline may develop in which the density increase with depth is due to vertical variations in salinity. In this case, the pycnocline is a *halocline*.

Figure 6.19 Density stratification in the ocean.

Do the layers look the same everywhere?

Remember: Temperature usually dominates ocean density

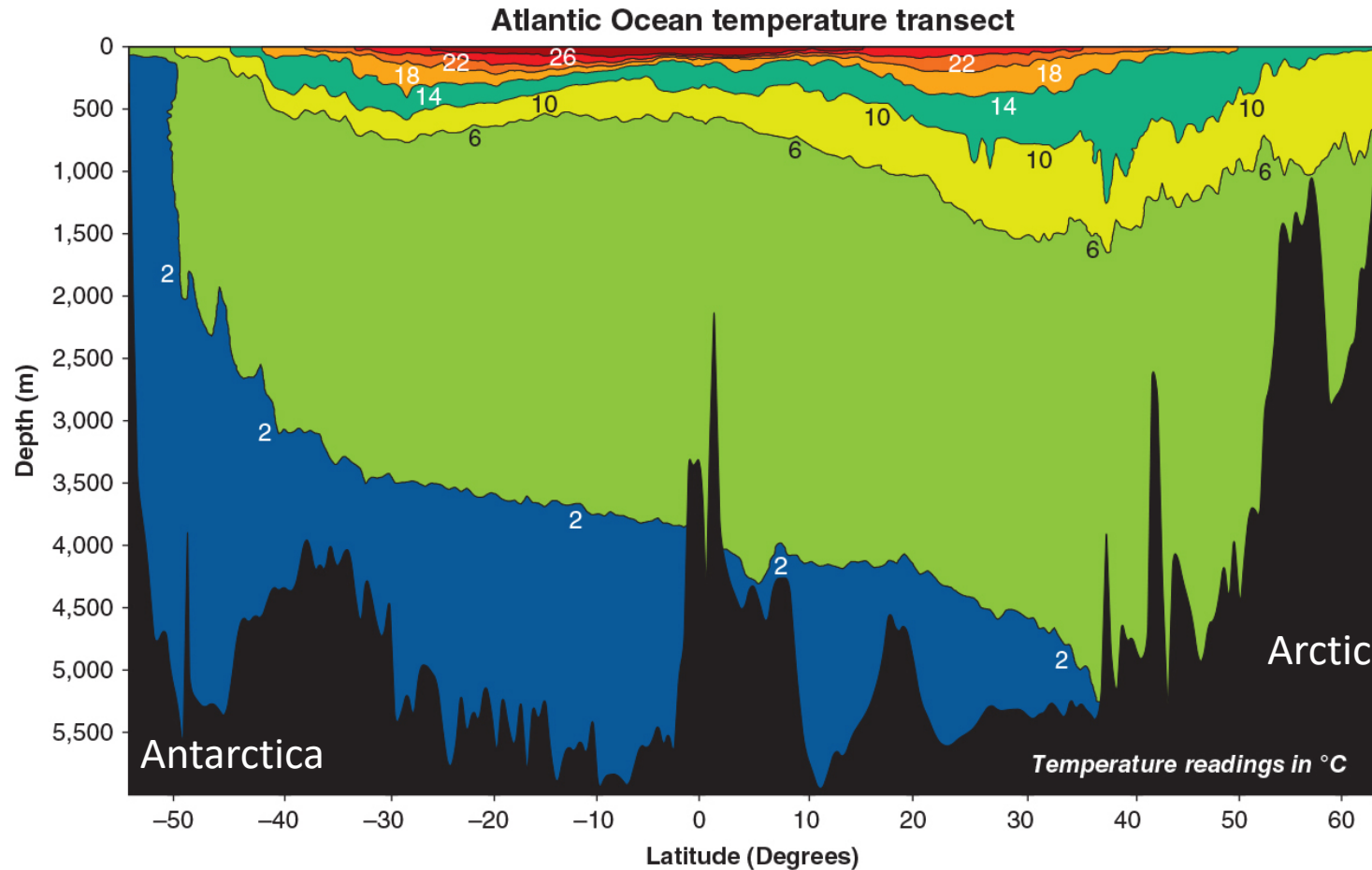


Figure 6.20 Temperature and salinity across the Atlantic Ocean basin at about 30° Degrees west longitude. Most seawater lies in cold darkness. In this image, a small and thin warm layer, concentrated near the equator, stands out above the relatively cool and consistently stable deep waters. As Earth warms, this layer will extend poleward. The irregular black shapes at the bottom of the graphs represent the ocean floor at the latitude shown.

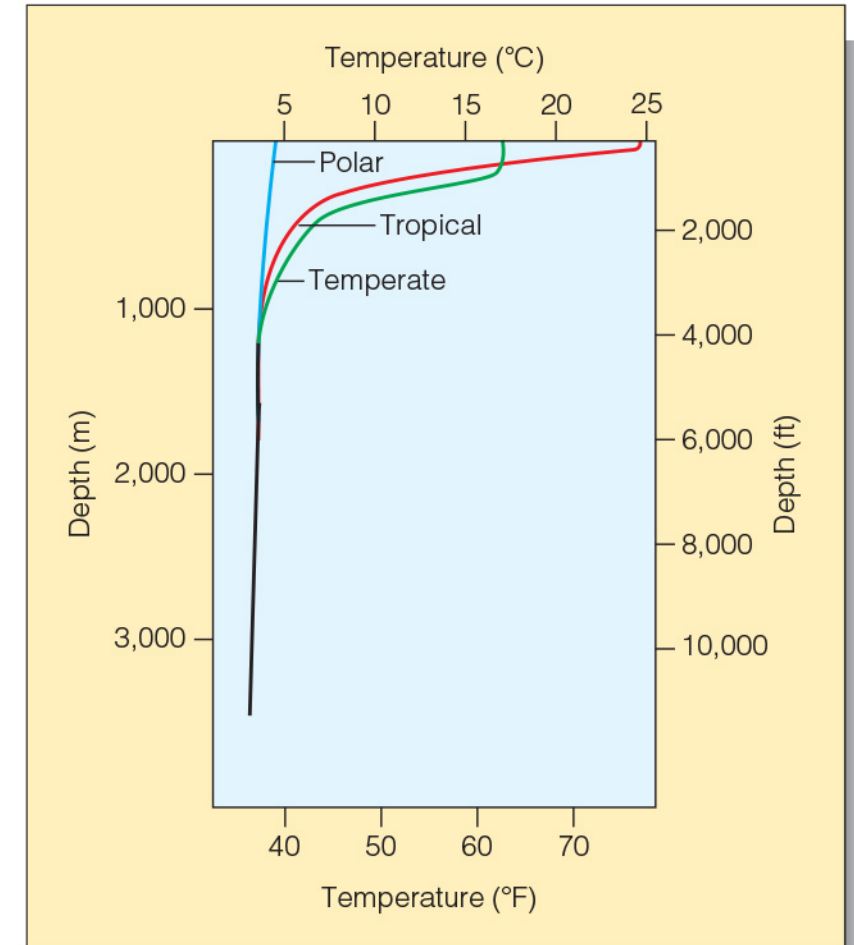


Figure 6.21 Typical temperature profiles at polar, tropical, and middle (temperate) latitudes. Note that polar waters lack a strong thermocline.